EXAM PACK

EEE421

Solid State Devices



Exam Pack

Session 2021–22 *Solutions and formula sheet of the term test (Tutorial 01); the final will be added*

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Prepared by Nur Mohammad

November 2026

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PAST PAPER SOLUTIONS

EEE421

Solid State Devices

Question Paper Solutions

Session 2021–22

Term test only: Tutorial 01 (Q7); the final examination is not yet held and will be added as Q1–Q6

QUESTION MAP

Q7

20 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

Prepared by Nur Mohammad

November 2026

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The paper at a glance

Q	Sub-part	Marks	Time	Pick
Tutorial 1 · 20 marks · 30 min				
Q7	(a) What the Fermi-Dirac Distribution Is	3	4 min	☐
	(b) Ionization Energy of Donors and Acceptors			
	(c) Equilibrium Distribution of Electrons and Holes in an Extrinsic Semiconductor	5	8 min	
	(d) Energy Above the Fermi Level for a 10 Percent Probability	5	8 min	
	(e) Electron Concentration in Si from the Fermi Level Position			
	(f) Silicon as a Dopant in Gallium Arsenide	7	10 min	

Formulas, symbols, equation numbers and constants follow Neamen, Semiconductor Physics and Devices (4th ed.); 7(f) is Neamen Problem 4.62.

Course Code: EEE-421

Title: Solid State Devices

Session 2021-22: term test only (Tutorial: 01); the final examination is not yet held

Session: 2021-22 **Tutorial: 01 only** **Final: not yet held**

Tutorial 1 (its questions 1(a) to 4 are 7(a) to 7(f))

[Tutorial: 01, Time: 30 minutes, Marks: 20 in all; the 3 marks of 7(a) cover 7(a) and 7(b) together, the 5 marks of 7(d) cover 7(d) and 7(e) together]

7. (a) What is Fermi Dirac distribution? 3

(b) Explain Ionization Energy.

(c) Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material. 5

(d) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%? 5

(e) Find the electron concentration (n_o) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature N_c =2.8 $\times$ 10^{19} cm^{-3})

(f) Silicon atoms, at a concentration of 7 $\times$ 10^{15} cm^{-3}, are added to gallium arsenide. Assume that the silicon atoms act as fully ionized dopant atoms and that 5 percent of the concentration added replaces gallium atoms and 95 percent replaces arsenic atoms. Let T = 300 K and GaAs has an intrinsic concentration of 1.8x10^6 cm $^{-3}$. 7

i) Determine the donor and acceptor concentration.
ii) Is the material n-type or p-type?
iii) Calculate the electron and hole concentration.
iv) Determine the position of the Fermi level with respect to E_{Fi}.

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7. (a) What is Fermi Dirac distribution? Marks: 3
   (b) Explain Ionization Energy.
   (c) Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material. Marks: 5
   (d) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%? Marks: 5
   (e) Find the electron concentration (n_o) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature N_c =2.8 $\times$ 10^{19} cm^{-3})
   (f) Silicon atoms, at a concentration of 7 $\times$ 10^{15} cm^{-3}, are added to gallium arsenide. Assume that the silicon atoms act as fully ionized dopant atoms and that 5 percent of the concentration added replaces gallium atoms and 95 percent replaces arsenic atoms. Let T = 300 K and GaAs has an intrinsic concentration of 1.8x10^6 cm $^{-3}$. Marks: 7
      i) Determine the donor and acceptor concentration.
      ii) Is the material n-type or p-type?
      iii) Calculate the electron and hole concentration.
      iv) Determine the position of the Fermi level with respect to E_{Fi}.

7.(a) What is Fermi Dirac distribution? Marks: 3

1. Theory

1. Energy states and electrons (Core Idea)

In a single atom, an electron can only have certain energies. In a crystal, billions of atoms sit close together, and each allowed energy spreads into a **band** of very many closely spaced allowed energies, called **quantum states**. For a semiconductor two bands matter:

- the **valence band**, with top edge E_v; at 0 K it is completely full (these are the bonding electrons);
- the **conduction band**, with bottom edge E_c; at 0 K it is completely empty.

Between them is the **band gap** $E_g = E_c - E_v$ (1.12 eV for Si), where no states exist.

A crystal has far more states than we can follow one by one, and the electrons keep exchanging energy with the vibrating lattice. So we do not ask "which state holds which electron?" but "**what is the probability** that a state at energy E holds an electron?" The rule that answers this is a **distribution function**.

2. Three statistical laws (Neamen, Section 3.5.1)

How particles share out the available states depends on what kind of particles they are:

- **Maxwell-Boltzmann:** the particles can be told apart (numbered), and any number may sit in one state. Example: gas molecules at low pressure.
- **Bose-Einstein:** the particles cannot be told apart, and any number may sit in one state. Example: photons.
- **Fermi-Dirac:** the particles cannot be told apart, and **at most one particle may sit in each quantum state** (the **Pauli exclusion principle**). Electrons in a crystal obey this law.

3. Where the Fermi-Dirac function comes from (Neamen, Section 3.5.2)

Take one energy level E_i with g_i quantum states and put N_i electrons into it ($N_i \le g_i$, one per state at most).

- The first electron can go into any of g_i states, the second into any of the $g_i - 1$ left, and so on. The number of ordered placements is

$$g_i(g_i - 1)\cdots(g_i - N_i + 1) = \frac{g_i!}{(g_i - N_i)!}$$

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(Neamen Eq. 3.76).

- Electrons cannot be told apart, so the $N_i!$ orderings of the same electrons among the same states are one arrangement. Dividing by $N_i!$:

$$W_i = \frac{g_i!}{N_i!\,(g_i - N_i)!}$$

(Eq. 3.77). Example: $g_i = 10$ states and $N_i = 9$ electrons give $W_i = 10$ (the one empty state can be in 10 places).

- For all levels together the number of ways is the product $W = \prod_i W_i$ (Eq. 3.78).

In equilibrium the system takes the **most probable** distribution: the set of N_i that makes W largest, while the total number of electrons and the total energy stay fixed. Carrying out this maximization gives (Neamen Eq. 3.79)

$$\frac{N(E)}{g(E)} = f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

where $N(E)$ is the number of electrons and $g(E)$ the number of quantum states per unit volume per unit energy. So $f_F(E)$ is the **fraction of the states at energy E that are filled**, which is the same as the **probability that one state at E is filled**.

4. The symbols

- E_F is the **Fermi energy** (Fermi level). It is not necessarily an allowed energy; it is a reference energy fixed by how many electrons the system holds.
- $k = 8.62 \times 10^{-5}$ eV/K is Boltzmann's constant and T the absolute temperature; $kT = 0.0259$ eV at 300 K. kT is the typical thermal energy of one particle.

5. What the function does

- **At $T = 0$ K:** for $E < E_F$ the exponential is $e^{-\infty} = 0$, so $f_F = 1$; for $E > E_F$ it is $e^{+\infty}$, so $f_F = 0$. Every state below E_F is full and every state above is empty: a sharp **step**.
- **At any $T > 0$:** at $E = E_F$ the exponential is $e^0 = 1$ and $f_F = 1/(1+1) = 1/2$. The Fermi level is the energy where a state is **half likely** to be filled.
- **Above zero temperature** the step rounds off over a few kT: some states above E_F are filled (electrons lifted by heat) and some below are empty.

$(E - E_F)/kT$	-3	-2	-1	0	$+1$	$+2$	$+3$	$+5$
$f_F(E)$	0.953	0.881	0.731	0.500	0.269	0.119	0.0474	0.0067

Table 1: Values of the Fermi-Dirac function a few kT from E_F

6. Empty states and symmetry

The probability that a state is **empty** is $1 - f_F(E)$. Since

$$1 - \frac{1}{1+e^x} = \frac{e^x}{1+e^x} = \frac{1}{1+e^{-x}}$$

with $x = (E - E_F)/kT$, the chance that a state ΔE **above** E_F is filled equals the chance that a state ΔE **below** E_F is empty (Neamen Fig. 3.34). This symmetry is why holes in the valence band behave like electrons in the conduction band.

7. The Boltzmann approximation

When $E - E_F \gg kT$, the exponential is much larger than 1, and the 1 in the denominator can be dropped (Neamen Eq. 3.80):

$$f_F(E) \approx \exp\left[-\frac{E - E_F}{kT}\right]$$

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At $E - E_F = 3kT$ this gives 0.0498 against the exact 0.0474, about 5 % high; further away it is better. It is used for electrons in the conduction band of a non-degenerate semiconductor, where $E_c - E_F$ is several kT.

2. Answer

Definition: The Fermi-Dirac distribution is the probability function that gives, for particles that obey the Pauli exclusion principle (electrons), the probability that an allowed quantum state at energy E is occupied by an electron in thermal equilibrium (Neamen Eq. 3.79):

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

where

- E_F is the Fermi energy (Fermi level)
- k is Boltzmann's constant and T the absolute temperature ($kT = 0.0259$ eV at 300 K)

Key Points:

- At $T = 0$ K, $f_F = 1$ for $E < E_F$ and $f_F = 0$ for $E > E_F$: all states up to E_F are full.
- For $T > 0$, $f_F(E_F) = 1/2$ at every temperature; the change from 1 to 0 is spread over a few kT around E_F, more widely when hotter.
- $1 - f_F(E)$ is the probability that the state is empty (a hole); f_F and $1 - f_F$ are symmetric about E_F.
- For $E - E_F \gg kT$ it reduces to the Boltzmann form $f_F \approx \exp[-(E - E_F)/kT]$.
- Multiplied by the density of states it gives the carrier distributions: $n(E) = g_c(E) f_F(E)$ and $p(E) = g_v(E)[1 - f_F(E)]$.

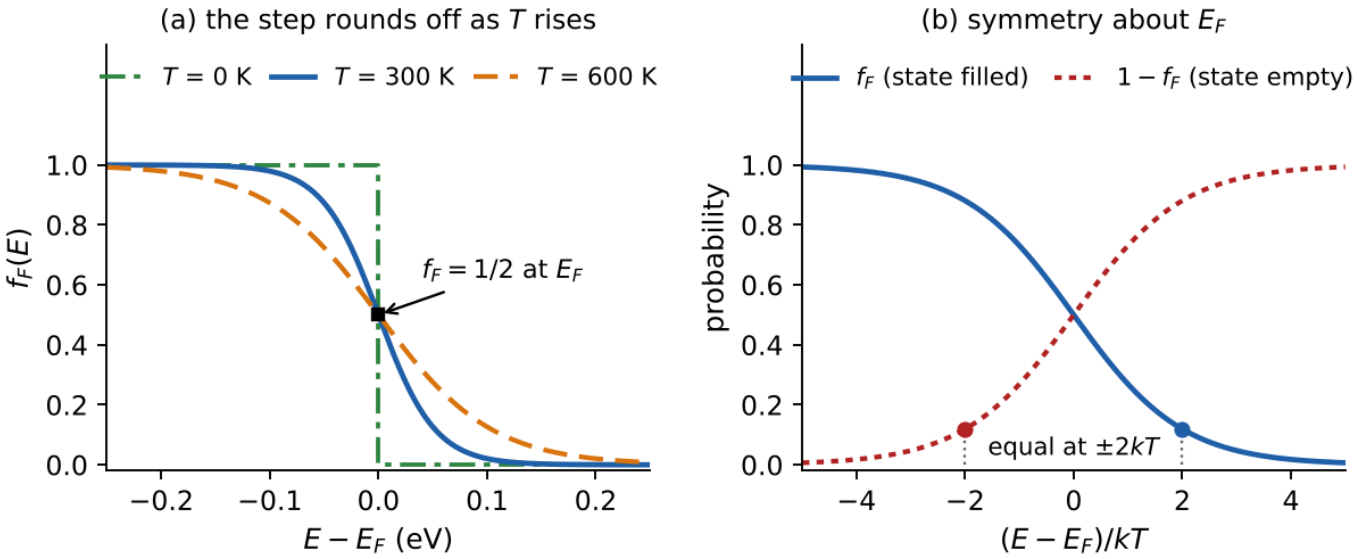


Figure 1: The Fermi-Dirac function at 0 K, 300 K and 600 K, and the symmetry of a filled and an empty state about the Fermi level

Reading the figure. In panel (a) the dash-dot green line is the sharp step at 0 K; the solid blue curve (300 K) and the dashed orange curve (600 K) all pass through $f_F = 1/2$ at E_F (black square), and the hotter curve spreads further. In panel (b) the solid blue curve is f_F and the dotted red curve is $1 - f_F$; the two dots at $\pm 2kT$ have the same height (0.119).

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Example (Neamen Example 3.6)

a state $3kT$ above E_F is filled with probability $1/(1+e^3) = 1/21.09 = 0.0474$, about 4.7 %.

3. Keyword Explanation

1. Quantum state

↩ back

One allowed solution for an electron (an energy, a wave vector and a spin); by the Pauli principle it holds at most one electron.

2. Pauli exclusion principle

↩ back

No two electrons can occupy the same quantum state.

3. Fermi energy

↩ back

The energy E_F at which the occupation probability is exactly 1/2. In equilibrium it is the same everywhere in a material.

4. Boltzmann approximation

↩ back

$f_F \approx \exp[-(E - E_F)/kT]$, valid when $E - E_F$ is more than about $3kT$.

4. Notes and Remarks

- **Marks:** the paper prints one mark, 3, beside its questions 1(a) and 1(b) together (here 7(a) and 7(b)); it is shown on 7(a).
- **Temperature dependence:** the Fermi function is a strong function of T. Neamen Example 3.7 finds the temperature (756 K) at which a state 0.30 eV below E_F is empty with 1 % probability.
- **Why not Maxwell-Boltzmann?** Classical statistics allows any number of electrons per state; then at 0 K every electron would fall into the lowest state, which is wrong. The "one per state" rule is what makes E_F and the step appear.
- The 10 % point of this function is worked out in 7(d), and the function is used to count carriers in 7(c).
- The 2020–21 paper uses the same function in its 1(a) and 2(d): ../2020-21/2020-21-term-final.md (extra reading).

7.(b) Explain Ionization Energy.

1. Theory

1. Ionization energy in general (Core Idea)

To **ionize** an atom means to remove an electron from it, leaving a positive ion. The **ionization energy** is the smallest energy that must be given to a bound electron to free it. For a hydrogen atom in its ground state it is 13.6 eV.

2. Donor atoms in silicon

Si has four valence electrons, and each Si atom forms four covalent bonds with its neighbours. Suppose a phosphorus (P) atom, from group V with **five** valence electrons, replaces a Si atom (a **substitutional impurity**):

- four of its electrons form the four bonds, like a Si atom;
- the **fifth** electron has no bond to join. It stays only loosely held by the P nucleus, which now has one positive charge left over (P^+).

At very low temperature this fifth electron circles the P^+ ion. A small energy frees it: it then moves through the crystal as a conduction electron, and the P atom is left as a fixed positive ion. Because the impurity "donates" an electron to the conduction band, it is called a **donor**. On the band diagram the bound fifth electron sits at a **donor level** E_d just below E_c, and the energy to free it is $E_c - E_d$.

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3. Acceptor atoms

A boron (B) atom, from group III with *three* valence electrons, can fill only three bonds; one bond is missing an electron. A valence electron from a neighbouring bond can move into this empty place with only a small energy; the B atom becomes a fixed negative ion (B^-) and a **hole** is left in the valence band. B "accepts" an electron and is called an **acceptor**. Its level E_a is just above E_v, and the energy needed is $E_a - E_v$.

4. The hydrogen model of a donor (Neamen, Section 4.2.2)

The fifth electron and the P^+ ion look like a hydrogen atom, with two changes: the electron moves inside silicon, so the Coulomb force is reduced by the **relative permittivity** ϵ_r ($\epsilon = \epsilon_r \epsilon_0$), and it moves with the **effective mass** m^* instead of m_0. The Bohr model then gives its energy step by step.

Step 1: force balance. The Coulomb attraction provides the centripetal force of a circular orbit of radius r_n and speed v (Neamen Eq. 4.27):

$$\frac{e^2}{4\pi\epsilon r_n^2} = \frac{m^* v^2}{r_n}$$

Step 2: quantized angular momentum. Bohr's condition (Eq. 4.28), with $n = 1, 2, 3, \ldots$:

$$m^* r_n v = n\hbar \quad \Rightarrow \quad v = \frac{n\hbar}{m^* r_n}$$

Step 3: the radius. Put this v into Step 1: $\dfrac{e^2}{4\pi\epsilon r_n^2} = \dfrac{m^*}{r_n} \cdot \dfrac{n^2\hbar^2}{m^{*2} r_n^2}$. Multiply both sides by r_n^3 and solve (Eq. 4.29):

$$r_n = \frac{n^2\hbar^2\, 4\pi\epsilon}{m^* e^2}$$

Divided by the Bohr radius $a_0 = 4\pi\epsilon_0\hbar^2/(m_0 e^2) = 0.53$ Å (Eq. 4.30), $r_n/a_0 = n^2\epsilon_r(m_0/m^*)$ (Eq. 4.31). For Si, $\epsilon_r = 11.7$ and $m^*/m_0 = 0.26$, so $r_1 = 45\, a_0 = 23.8$ Å, about four lattice constants: the orbit encloses many Si atoms, so the electron is only weakly bound.

Step 4: kinetic energy. $T = \frac{1}{2} m^* v^2$. From Step 1, $m^* v^2 = e^2/(4\pi\epsilon r_n)$, so $T = e^2/(8\pi\epsilon r_n)$. With r_n from Step 3 (Eq. 4.35):

$$T = \frac{m^* e^4}{2(n\hbar)^2 (4\pi\epsilon)^2}$$

Step 5: potential energy. For a charge $-e$ at distance r_n from $+e$ (Eq. 4.36):

$$V = -\frac{e^2}{4\pi\epsilon r_n} = -\frac{m^* e^4}{(n\hbar)^2 (4\pi\epsilon)^2}$$

Step 6: total energy. Adding (Eq. 4.37):

$$E = T + V = -\frac{m^* e^4}{2(n\hbar)^2 (4\pi\epsilon)^2}$$

The minus sign means the electron is bound. The **ionization energy** is the energy needed to bring it from $n = 1$ to $E = 0$ (free), i.e. $|E_1|$.

Step 7: scale from hydrogen. For hydrogen ($m^* = m_0$, $\epsilon = \epsilon_0$) $|E_1| = 13.6$ eV. The formula has m^* on top and ϵ^2 below, so

$$E_{ion} = 13.6 \text{ eV} \times \frac{m^*/m_0}{\epsilon_r^2} = 13.6 \times \frac{0.26}{(11.7)^2} = 0.0258 \text{ eV} = 25.8 \text{ meV}$$

This is Neamen's value for Si: tiny compared with $E_g = 1.12$ eV, and about equal to $kT = 25.9$ meV at room temperature.

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5. Measured values (Neamen Table 4.3)

Impurity (in Si)	Type	Ionization energy (eV)
Phosphorus	donor	0.045
Arsenic	donor	0.05
Boron	acceptor	0.045
Aluminum	acceptor	0.06

Table 2: Measured ionization energies of dopants in Si (Neamen Table 4.3)

The simple model (0.026 eV) gives the right size; the real values are somewhat larger because close to the ion the crystal is not a smooth dielectric.

2. Answer

Definition: The ionization energy of an impurity (dopant) atom is the energy needed to free its loosely bound extra electron (donor) or to bring a valence electron into its empty bond (acceptor), i.e. to turn the neutral impurity into a fixed ion and create a free carrier.

- **Donor** (P, As in Si): ionization energy $= E_c - E_d$, the energy to lift the fifth electron from the donor level E_d into the conduction band. The donor becomes N_d^+ and a free electron appears.
- **Acceptor** (B, Al in Si): ionization energy $= E_a - E_v$, the energy to lift a valence electron to the acceptor level E_a. The acceptor becomes N_a^- and a free hole appears.

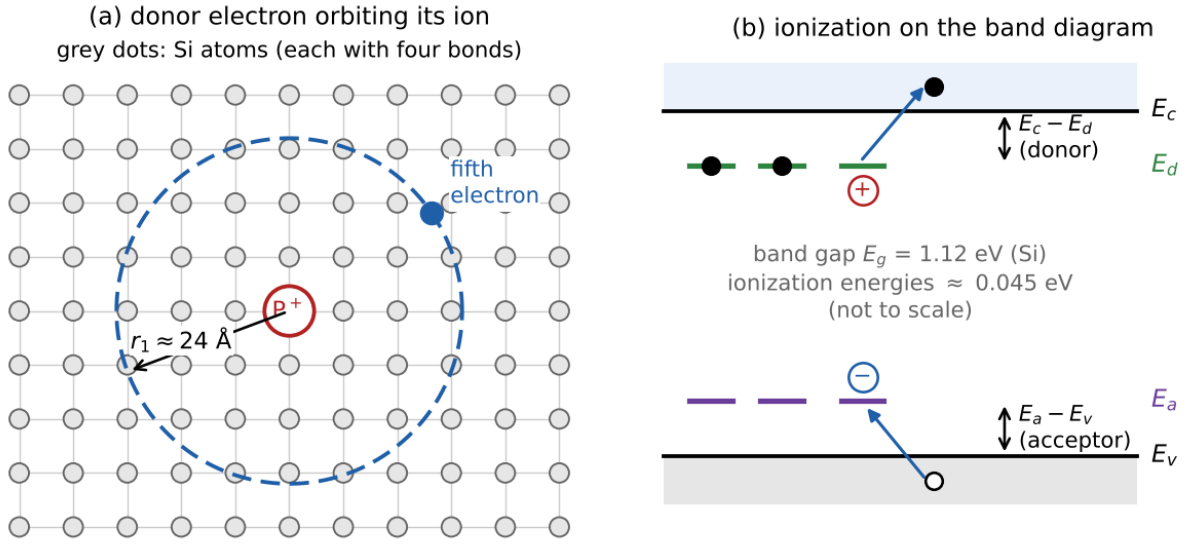


Figure 2: The hydrogen-like orbit of a donor electron in Si, and donor and acceptor ionization on the band diagram

Reading the figure. Panel (a): the fifth electron (blue dot) of a P donor circles the P^+ ion on a dashed orbit of radius about 24 Å, around many Si atoms (grey dots), so it is weakly bound. Panel (b): two donors at E_d (solid green lines) still hold their electrons (black dots); the third has given its electron to the conduction band (blue arrow) and is a positive ion (red +). Below, a valence electron moves up to an acceptor level E_a (solid purple line), leaving a hole (open circle) and a negative acceptor ion (blue −). The gap between the levels and the band edges is the ionization energy.

Estimate (hydrogen model, Theory 4):

$$E_{ion} = 13.6 \text{ eV} \times \frac{m^*/m_0}{\epsilon_r^2} = 13.6 \times \frac{0.26}{11.7^2} = 25.8 \text{ meV}$$

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against measured 0.045 eV (P) and 0.05 eV (As) in Si.

Why it matters: the ionization energy (about 0.05 eV) is comparable to $kT = 0.0259$ eV and very small compared with $E_g = 1.12$ eV. So at room temperature almost every dopant is ionized (**complete ionization**), and $n_0 \approx N_d$ or $p_0 \approx N_a$. At very low temperature ($kT \ll E_c - E_d$) the electrons stay on the donors (**freeze-out**).

3. Keyword Explanation

1. Donor

↩ back

An impurity that gives an electron to the conduction band (group V in Si), becoming a fixed positive ion.

2. Acceptor

↩ back

An impurity that takes an electron from the valence band (group III in Si), creating a hole and becoming a fixed negative ion.

3. Effective mass

↩ back

The mass m^* that makes an electron in the crystal obey Newton's law; it includes the effect of the lattice.

4. Complete ionization

↩ back

All dopant atoms are ionized: $N_d^+ = N_d$, $N_a^- = N_a$.

4. Notes and Remarks

- **Marks:** 7(b) has no mark of its own; the 3 marks shown on 7(a) are for 7(a) and 7(b) together, as printed.
- **Book check:** with the same model Neamen's exercise TYU 4.7 gives 5.30 meV for GaAs ($\epsilon_r = 13.1$, $m^* = 0.067m_0$) and 6.37 meV for Ge; the calculation script gives 5.31 meV and 6.38 meV.
- **Atom versus dopant:** "ionization energy" of a free atom (13.6 eV for H) is much larger; inside the crystal the screening by ϵ_r and the small m^* reduce it by $\epsilon_r^2/(m^*/m_0) \approx 500$ times.
- **III-V crystals:** in GaAs a group IV atom such as Si is a donor on a Ga site and an acceptor on an As site; this is used in 7(f).
- How far ionization is complete at 300 K is worked out in the 2020–21 paper, 1(d) and 6(b) (extra reading: ../2020-21/2020-21-term-final.md).

7.(c) Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material. Marks: 5

1. Theory

1. Intrinsic and extrinsic semiconductors (Core Idea)

- An **intrinsic** semiconductor is a pure crystal. Every electron in the conduction band left a hole in the valence band, so $n_0 = p_0 = n_i$, the **intrinsic carrier concentration** (1.5×10^{10} cm^{-3} for Si at 300 K). Its Fermi level is the **intrinsic Fermi level** E_{Fi}, close to midgap.
- An **extrinsic** semiconductor has dopant atoms added on purpose (7(b)): donors make it **n-type** ($n_0 > p_0$), acceptors make it **p-type** ($p_0 > n_0$).

The subscript 0 means **thermal equilibrium**: no light, no applied voltage, a steady temperature.

Doping changes the number of electrons in the crystal, so the Fermi level moves: up towards E_c in n-type and down towards E_v in p-type. The question asks for the formulas that give n_0 and p_0 in terms of where E_F is. Neamen calls them the "equilibrium distribution of electrons and holes" (Section 4.3.1).

2. Number of carriers = number of states × probability

The number of electrons per unit volume with energy between E and $E + dE$ is the number of states there times the probability that each is filled (Neamen Eq. 4.1, 4.2):

$$n(E) = g_c(E) f_F(E), \qquad p(E) = g_v(E)\,[1 - f_F(E)]$$

where f_F is the Fermi-Dirac function (7(a)) and $1 - f_F$ the probability that a state is empty (a hole).

3. The density of states (background for Step 4)

$g_c(E)$ is the number of allowed electron states per unit volume per unit energy near E. It is found by counting (Neamen, Section 3.4):

- An electron in a cube of side a has standing-wave solutions with $k_x = n_x\pi/a$, $k_y = n_y\pi/a$, $k_z = n_z\pi/a$ (n_x, n_y, n_z positive integers). So in k-space each state takes a volume $(\pi/a)^3$ (Eq. 3.62).
- The states with $|k|$ between k and $k+dk$ fill a spherical shell of volume $4\pi k^2 dk$. Only the positive octant counts ($\frac{1}{8}$), and each k holds two electrons (two spins, factor 2):

$$g_T(k)\,dk = 2 \cdot \frac{1}{8} \cdot \frac{4\pi k^2\,dk}{(\pi/a)^3} = \frac{\pi k^2\,dk}{\pi^3}\,a^3$$

(Eq. 3.63, 3.64).

- Near the bottom of the conduction band an electron behaves like a free particle of mass m_n^*: $E - E_c = \hbar^2 k^2/(2m_n^*)$ (Eq. 3.71). So $k^2 = 2m_n^*(E - E_c)/\hbar^2$ and $dk = \dfrac{1}{\hbar}\sqrt{\dfrac{m_n^*}{2(E - E_c)}}\,dE$.
- Substituting, using $\hbar = h/2\pi$, and dividing by the volume a^3 (Eq. 3.67 to 3.69):

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c} \quad (E \geq E_c), \qquad g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E} \quad (E \leq E_v)$$

(Neamen Eq. 3.72 and 3.75). There are no states at the band edge itself, and more and more states higher in the band.

4. The picture

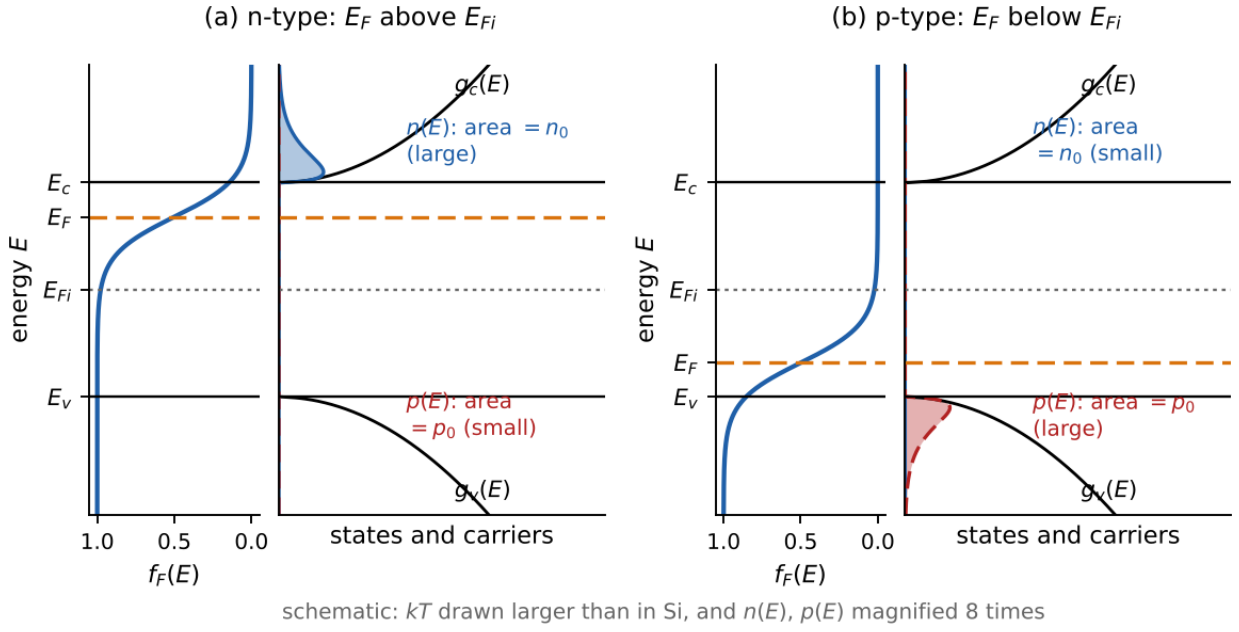


Figure 3: Density of states, Fermi function and the electron and hole distributions for n-type and p-type material

Reading the figure. In each panel the left plot is $f_F(E)$ (solid blue) and the right plot the densities of states g_c and g_v (solid black). Their product gives the shaded $n(E)$ (blue, above E_c) and $p(E)$ (red,

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dashed outline, below E_v); the area under each is n_0 or p_0. In panel (a) E_F (dashed orange) is above E_{Fi} (dotted grey): the tail of f_F reaches into the conduction band, so n_0 is large and p_0 very small (n-type). In panel (b) E_F is below E_{Fi} and the situation is reversed (p-type). Same idea as Neamen Fig. 4.8 and 4.9; the drawing is schematic.

5. Two tools used in the derivation

- **Non-degenerate semiconductor:** E_F lies more than about $3kT$ inside the gap from both band edges, so the Boltzmann approximation (7(a), Theory 7) holds for every conduction band state and every valence band state.
- **The integral** $\int_0^\infty \eta^{1/2} e^{-\eta}\, d\eta = \sqrt{\pi}/2$, worked out in Step 6 of the Answer.

2. Answer

The derivation follows Neamen, Sections 4.1 and 4.3 (Eq. 4.1 to 4.43). Steps marked **(Basic)** work out a general mathematical tool; a reader who knows it may skip them.

Step 1: Carriers per unit energy

Number of electrons in dE = (number of states in dE) × (probability each is filled); number of holes = states × probability each is empty:

$$n(E) = g_c(E) f_F(E), \qquad p(E) = g_v(E)\,[1 - f_F(E)] \qquad \text{(Neamen 4.1, 4.2)}$$

Step 2: Total electron concentration

Add up $n(E)$ over the whole conduction band. The upper limit is taken as ∞, because f_F is already practically zero a few tenths of an eV above E_c:

$$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE \qquad (4.3)$$

Step 3: Boltzmann approximation for the conduction band

For a non-degenerate semiconductor, every conduction band state has $E - E_F \geq E_c - E_F \gg kT$, so

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)} \approx \exp\left[\frac{-(E - E_F)}{kT}\right] \qquad (4.4)$$

Step 4: Substitute the density of states

With $g_c(E)$ from Theory 3:

$$n_0 = \frac{4\pi (2m_n^*)^{3/2}}{h^3} \int_{E_c}^{\infty} \sqrt{E - E_c}\, \exp\left[\frac{-(E - E_F)}{kT}\right] dE \qquad (4.5)$$

Step 5: Change of variable

Let $\eta = (E - E_c)/kT$ (Eq. 4.6). Then $E - E_c = kT\eta$, $dE = kT\, d\eta$, and the limits $E_c \to \infty$ become $0 \to \infty$. Split the exponent by adding and subtracting E_c:

$$E - E_F = (E - E_c) + (E_c - E_F) = kT\eta + (E_c - E_F)$$

so $\exp[-(E - E_F)/kT] = \exp[-(E_c - E_F)/kT]\, e^{-\eta}$. The first factor does not depend on E and comes out of the integral. Also $\sqrt{E - E_c}\, dE = (kT)^{1/2}\eta^{1/2} \cdot kT\, d\eta = (kT)^{3/2}\eta^{1/2} d\eta$. Hence

$$n_0 = \frac{4\pi (2m_n^* kT)^{3/2}}{h^3} \exp\left[\frac{-(E_c - E_F)}{kT}\right] \int_0^{\infty} \eta^{1/2} e^{-\eta}\, d\eta \qquad (4.7)$$

Step 6 (Basic): The integral $\int_0^\infty \eta^{1/2} e^{-\eta} d\eta = \sqrt{\pi}/2$

Put $\eta = x^2$, $d\eta = 2x\, dx$: the integral becomes $\int_0^\infty x\, e^{-x^2}\, 2x\, dx = \int_0^\infty x \cdot 2x e^{-x^2} dx$.

Integrate by parts with $u = x$ and $dv = 2xe^{-x^2} dx$, so $du = dx$ and $v = -e^{-x^2}$:

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$$\int_0^\infty x \cdot 2xe^{-x^2}dx = \Big[-xe^{-x^2}\Big]_0^\infty + \int_0^\infty e^{-x^2}dx = 0 + G$$

The Gaussian integral $G = \int_0^\infty e^{-x^2}dx$ is found by squaring it and using polar coordinates over the first quadrant ($x^2 + y^2 = r^2$, $dx\,dy = r\,dr\,d\theta$):

$$G^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)}dx\,dy = \int_0^{\pi/2}\int_0^\infty e^{-r^2} r\,dr\,d\theta = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}$$

so $G = \sqrt{\pi}/2$ and

$$\int_0^\infty \eta^{1/2}e^{-\eta}\,d\eta = \frac{\sqrt{\pi}}{2} \qquad (4.8)$$

Step 7: Electron concentration and N_c

Put Step 6 into Step 5:

$$n_0 = \frac{4\pi(2m_n^*kT)^{3/2}}{h^3}\cdot\frac{\sqrt{\pi}}{2}\exp\left[\frac{-(E_c-E_F)}{kT}\right] = \frac{2\pi^{3/2}(2m_n^*kT)^{3/2}}{h^3}\exp\left[\frac{-(E_c-E_F)}{kT}\right]$$

Since $\pi^{3/2}(2m_n^*kT)^{3/2}/h^3 = (2\pi m_n^*kT/h^2)^{3/2}$, this is (Eq. 4.9 to 4.11)

$$n_0 = N_c\exp\left[\frac{-(E_c-E_F)}{kT}\right], \qquad N_c = 2\left(\frac{2\pi m_n^*kT}{h^2}\right)^{3/2}$$

N_c is the **effective density of states** of the conduction band (2.8×10^{19} cm^{-3} for Si at 300 K): the band acts as if N_c states sat at E_c, each filled with the Boltzmann probability.

Step 8: Hole concentration

The same steps for the valence band. Total holes (Eq. 4.12):

$$p_0 = \int_{-\infty}^{E_v} g_v(E)\,[1 - f_F(E)]\,dE$$

The empty probability is $1 - f_F = 1/[1+\exp((E_F-E)/kT)]$ (Theory 6 of 7(a)); for $E_F - E \gg kT$ it is $\approx \exp[-(E_F-E)/kT]$ (Eq. 4.13). Let $\eta' = (E_v - E)/kT$ (Eq. 4.15), so $E_F - E = (E_F - E_v) + kT\eta'$, $\sqrt{E_v - E} = (kT)^{1/2}\eta'^{1/2}$ and $dE = -kT\,d\eta'$; the limits $E = -\infty \to E_v$ become $\eta' = \infty \to 0$, and the minus sign of dE turns them back to $0 \to \infty$:

$$p_0 = \frac{4\pi(2m_p^*kT)^{3/2}}{h^3}\exp\left[\frac{-(E_F-E_v)}{kT}\right]\int_0^\infty \eta'^{1/2}e^{-\eta'}\,d\eta'$$

With Step 6 and the same algebra as Step 7 (Eq. 4.17 to 4.19):

$$p_0 = N_v\exp\left[\frac{-(E_F-E_v)}{kT}\right], \qquad N_v = 2\left(\frac{2\pi m_p^*kT}{h^2}\right)^{3/2}$$

($N_v = 1.04\times10^{19}$ cm^{-3} for Si at 300 K). These two results hold for any position of E_F, intrinsic or extrinsic, as long as the material is non-degenerate.

Step 9: The intrinsic reference

In intrinsic material $E_F = E_{Fi}$ and $n_0 = p_0 = n_i$, so Steps 7 and 8 give (Eq. 4.20, 4.21)

$$n_i = N_c\exp\left[\frac{-(E_c-E_{Fi})}{kT}\right], \qquad n_i = N_v\exp\left[\frac{-(E_{Fi}-E_v)}{kT}\right]$$

Step 10: Electrons and holes measured from E_{Fi}

Add and subtract E_{Fi} in the exponent of Step 7 (Eq. 4.38):

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$$n_0 = N_c \exp\left[\frac{-(E_c - E_{Fi}) + (E_F - E_{Fi})}{kT}\right] = N_c \exp\left[\frac{-(E_c - E_{Fi})}{kT}\right] \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$$

The first factor is n_i (Step 9), so (Eq. 4.39)

$$\boxed{n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]}$$

In the same way, writing $E_F - E_v = (E_F - E_{Fi}) + (E_{Fi} - E_v)$ in Step 8 and using the second form of Step 9 (Eq. 4.40):

$$\boxed{p_0 = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]}$$

Step 11: The $n_0 p_0$ product

Multiply Steps 7 and 8; the E_F terms cancel because $(E_c - E_F) + (E_F - E_v) = E_c - E_v = E_g$ (Eq. 4.41, 4.42):

$$n_0 p_0 = N_c N_v \exp\left[\frac{-(E_c - E_F)}{kT}\right] \exp\left[\frac{-(E_F - E_v)}{kT}\right] = N_c N_v \exp\left[\frac{-E_g}{kT}\right]$$

This is the same as n_i^2 from Step 9 (multiply its two forms), so (Eq. 4.43)

$$\boxed{n_0 p_0 = n_i^2}$$

for any non-degenerate semiconductor in equilibrium, doped or not (the **mass-action law**).

Step 12: What the equations say for extrinsic material

- **n-type** (donors): $E_F > E_{Fi}$, so by Step 10 $n_0 > n_i > p_0$. With complete ionization and $N_d \gg n_i$, $n_0 \approx N_d$, and Step 10 and Step 7 turn into the Fermi level position:

$$E_F - E_{Fi} = kT \ln \frac{N_d}{n_i}, \qquad E_c - E_F = kT \ln \frac{N_c}{N_d} \qquad \text{(Neamen 4.65, 4.64)}$$

- **p-type** (acceptors): $E_F < E_{Fi}$, so $p_0 > n_i > n_0$; with $p_0 \approx N_a$:

$$E_{Fi} - E_F = kT \ln \frac{N_a}{n_i}, \qquad E_F - E_v = kT \ln \frac{N_v}{N_a} \qquad (4.68,\ 4.67)$$

Example (Si, 300 K, $N_d = 10^{16}$ cm^{-3})

$n_0 = 10^{16}$ cm^{-3}, $p_0 = n_i^2/n_0 = (1.5 \times 10^{10})^2/10^{16} = 2.25 \times 10^4$ cm^{-3}, $E_F - E_{Fi} = 0.0259 \ln(10^{16}/1.5 \times 10^{10}) = 0.347$ eV and $E_c - E_F = 0.0259 \ln(2.8 \times 10^{19}/10^{16}) = 0.206$ eV.

Final Answer:

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right] = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$$

$$p_0 = N_v \exp\left[\frac{-(E_F - E_v)}{kT}\right] = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]$$

$$n_0 p_0 = n_i^2$$

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Steps to remember

$n(E) = g_c f_F \to$ integrate over the band $\to$ Boltzmann $f_F \to$ put in $g_c \propto \sqrt{E - E_c} \to \eta = (E - E_c)/kT \to \int \eta^{1/2} e^{-\eta} = \sqrt{\pi}/2 \to n_0 = N_c e^{-(E_c - E_F)/kT} \to$ same for holes $\to$ add and subtract $E_{Fi} \to n_0 = n_i e^{(E_F - E_{Fi})/kT}$, $p_0 = n_i e^{-(E_F - E_{Fi})/kT} \to$ multiply: $n_0 p_0 = n_i^2$.

3. Keyword Explanation

1. Extrinsic semiconductor

↩ back

A semiconductor with controlled amounts of donor or acceptor impurities, so that $n_0 \neq p_0$.

2. Effective density of states

↩ back

N_c (N_v): the band's states lumped at the band edge; $n_0 = N_c \times$ (Boltzmann probability at E_c).

3. Intrinsic Fermi level

↩ back

The Fermi level E_{Fi} of the undoped crystal; in Si it is 0.013 eV below midgap.

4. Non-degenerate

↩ back

E_F more than a few kT inside the gap, so the Boltzmann approximation and $n_0 p_0 = n_i^2$ hold.

5. Majority and minority carriers

↩ back

The more numerous carrier (electrons in n-type) and the less numerous one (holes in n-type).

4. Notes and Remarks

- **Validity:** every result here assumes the Boltzmann approximation. For a degenerate semiconductor (E_F within about $3kT$ of a band edge, doping above about 10^{18} cm^{-3} in Si) the Fermi-Dirac integral (Neamen 4.3.3) must be used, and $n_0 p_0 = n_i^2$ no longer holds.
- **Charge neutrality** decides where E_F lies for given doping: $n_0 + N_a^- = p_0 + N_d^+$ (Neamen 4.56). It is used in 7(f), where both donors and acceptors are present.
- **Book example:** Neamen Example 4.5 uses Step 7 and Step 8 with $E_c - E_F = 0.25$ eV in Si and gets $n_0 = 1.8 \times 10^{15}$ and $p_0 = 2.7 \times 10^4$ cm^{-3}; the script gives 1.80×10^{15} and 2.68×10^4 cm^{-3}.
- **A small inconsistency in the book's constants:** $\sqrt{N_c N_v}\exp(-E_g/2kT) = 6.95 \times 10^9$ cm^{-3} for Si, not the accepted 1.5×10^{10} cm^{-3} (Neamen notes this). So a Fermi level found through N_c and one found through n_i differ by about 0.02 eV; use one route for one problem.
- The same derivation underlies the 2020–21 paper's 1(a) (extra reading: ../2020-21/2020-21-term-final.md).

7.(d) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%? Marks: 5

1. Theory

1. The basics

Electrons in a solid can only sit in allowed states, at most one per state. In thermal equilibrium the probability that a state at energy E holds an electron is the Fermi-Dirac function (7(a)):

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

where E_F is the Fermi level and kT the thermal energy (0.0259 eV at 300 K).

2. Why "in units of kT"

The function depends on E only through $u = (E - E_F)/kT$: $f = 1/(1 + e^u)$. So the answer "how far above E_F" in units of kT is the same for every material and every temperature; only its value in eV depends on T.

3. What to expect

At E_F ($u = 0$) the probability is 0.5; at $u = 2$ it is 0.119 and at $u = 3$ it is 0.047 (7(a), Theory 5). So 10 % lies a little above $2kT$.

4. Inverting the function

Solving $f = 1/(1 + e^u)$ for u: $1 + e^u = 1/f$, $e^u = 1/f - 1$, so

$$u = \frac{E - E_F}{kT} = \ln\left(\frac{1}{f} - 1\right)$$

2. Answer

Given: $f_F(E) = 10\,\% = 0.10$; find $(E - E_F)$ in units of kT.

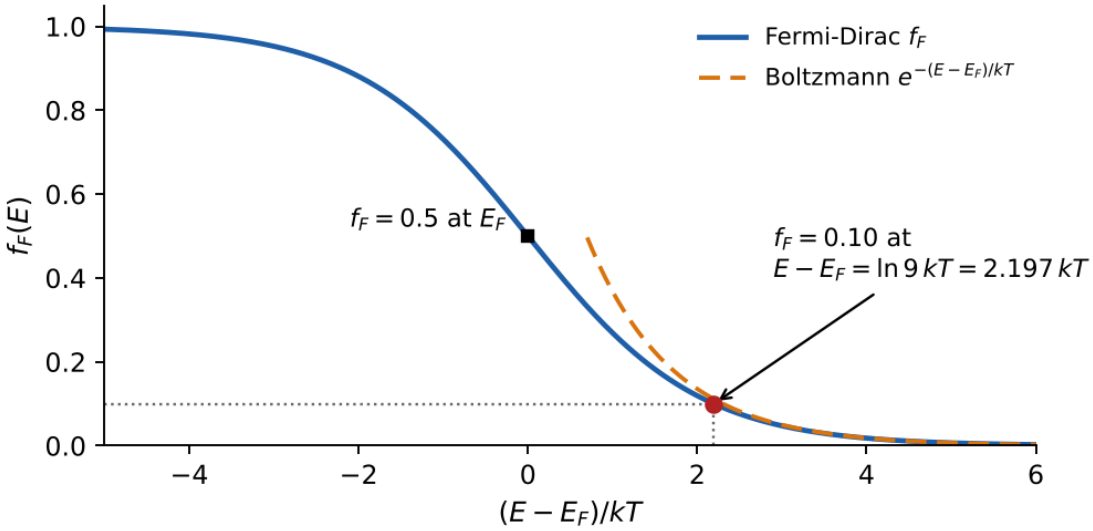


Figure 4: The Fermi-Dirac function against energy in units of kT, with the 10 percent point

Reading the figure. The solid blue curve is the Fermi-Dirac function; it is 0.5 at E_F (black square) and 0.10 at $2.197\,kT$ (red dot, dotted grey guide lines). The dashed orange curve is the Boltzmann approximation; near $2.2\,kT$ it is still visibly above the exact curve.

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Step 1: Set the Fermi function to 0.10

$$\frac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)} = 0.10$$

Step 2: Solve for the exponential

Take the reciprocal of both sides:

$$1+\exp\left(\frac{E-E_F}{kT}\right) = \frac{1}{0.10} = 10 \quad \Rightarrow \quad \exp\left(\frac{E-E_F}{kT}\right) = 9$$

Step 3: Take the natural logarithm

$$\frac{E-E_F}{kT} = \ln 9 = 2.197$$

$$\boxed{E - E_F = 2.197\,kT \approx 2.2\,kT}$$

At $T = 300$ K this is $2.197 \times 0.0259 = 0.0569$ eV above the Fermi level. So a state only about $2.2\,kT$ above E_F is already filled just one time in ten.

Final Answer: $E = E_F + kT\ln 9 = E_F + 2.197\,kT$ (0.0569 eV above E_F at 300 K).

3. Keyword Explanation

1. Occupation probability

↩ back

The chance $f_F(E)$ that a quantum state at energy E holds an electron.

2. Thermal energy kT

↩ back

The natural energy unit of thermal equilibrium; 0.0259 eV at 300 K.

4. Notes and Remarks

- **Marks:** the paper prints one mark, 5, beside its questions 3(a) and 3(b) together (here 7(d) and 7(e)); it is shown on 7(d).
- **Do not use the Boltzmann form here:** $e^{-u} = 0.10$ gives $u = \ln 10 = 2.303$, 4.8 % too far; at that point the exact probability is only $1/11 = 0.091$. The approximation needs $E - E_F \gtrsim 3kT$.
- **Holes:** by the symmetry of 7(a), a state $2.197\,kT$ **below** E_F is **empty** with probability 10 %.
- **Repeated question:** the same question is 2(d)(i) of the 2020–21 paper (extra reading: ../2020-21/2020-21-term-final.md).

7.(e) Find the electron concentration (n_o) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature N_c =2.8 × 10^{19} cm^{-3})

1. Theory

1. The basics

The free electrons of a semiconductor are in the conduction band, above E_c. Their number per cm^3, n_0, is the number of conduction band states times the probability that each is filled, added over the band. The probability is the Fermi-Dirac function of 7(a); the closer E_F is to E_c, the higher the probability and the more electrons.

2. The formula

When $E_c - E_F$ is more than about $3kT$ (non-degenerate), the sum gives (derived in full in 7(c), Steps 1 to 7; Neamen Eq. 4.11)

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

where

- N_c is the **effective density of states** of the conduction band: the band behaves as if N_c states per cm^3 sat at E_c. For Si at 300 K, $N_c = 2.8 \times 10^{19}$ cm^{-3}
- $\exp[-(E_c - E_F)/kT]$ is the (Boltzmann) probability that a state at E_c is filled

3. How sensitive it is

Every kT (0.0259 eV) that E_F moves down from E_c divides n_0 by $e = 2.718$; every $kT \ln 10 = 0.0596$ eV divides it by 10.

2. Answer

Given:

- Si at room temperature, $T = 300$ K, $kT = 0.0259$ eV
- $E_c - E_F = 0.5$ eV
- $N_c = 2.8 \times 10^{19}$ cm^{-3}

Formula (Neamen Eq. 4.11):

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Check of the approximation: $E_c - E_F = 0.5/0.0259 = 19.3\,kT \gg 3kT$, so the Boltzmann form is accurate (the exact Fermi factor at E_c agrees to better than 1 part in 10^8).

Step 1: The exponent

$$\frac{E_c - E_F}{kT} = \frac{0.5}{0.0259} = 19.305, \qquad \exp(-19.305) = 4.130 \times 10^{-9}$$

This is the probability that a state at E_c holds an electron.

Step 2: The concentration

$$n_0 = (2.8 \times 10^{19})(4.130 \times 10^{-9})$$

$$\boxed{n_0 = 1.16 \times 10^{11} \text{ cm}^{-3}}$$

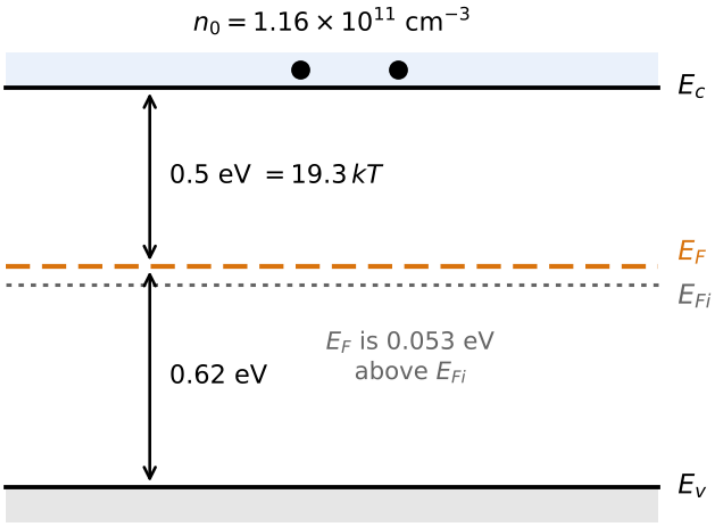


Figure 5: Band diagram of the Si sample at 300 K, to scale

Reading the figure. The solid black lines are E_c and E_v, 1.12 eV apart. The dashed orange E_F is 0.5 eV (19.3 kT) below E_c and 0.62 eV above E_v, just above the dotted grey intrinsic level E_{Fi}.

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Meaning: n_0 is about 7.7 times $n_i = 1.5 \times 10^{10}$ cm^{-3}, so the Si is very lightly n-type; E_F is $0.0259 \ln(1.16 \times 10^{11}/1.5 \times 10^{10}) = 0.053$ eV above E_{Fi}.

Final Answer: $n_0 = 2.8 \times 10^{19} \exp(-0.5/0.0259) = 1.16 \times 10^{11}$ cm^{-3}.

3. Keyword Explanation

1. Effective density of states

↩ back

N_c: the number of states that, placed at E_c, would hold the same number of electrons as the whole conduction band.

2. Room temperature

↩ back

Taken as $T = 300$ K, where $kT = 0.0259$ eV.

4. Notes and Remarks

- **Marks:** 7(e) has no mark of its own; the 5 marks shown on 7(d) are for 7(d) and 7(e) together, as printed.
- **Notation:** the paper writes the electron concentration as "(no)", read as n_o, the book's n_0.
- **Holes:** $p_0 = n_i^2/n_0 = (1.5 \times 10^{10})^2/(1.16 \times 10^{11}) = 1.95 \times 10^9$ cm^{-3}. Through N_v, $p_0 = 1.04 \times 10^{19} \exp(-0.62/0.0259) = 4.18 \times 10^8$ cm^{-3}; the two differ because Neamen's N_c, N_v and E_g give $n_i = 6.95 \times 10^9$, not 1.5×10^{10} cm^{-3} (see 7(c), Notes).
- **Book example:** Neamen Example 4.5 is the same calculation with $E_c - E_F = 0.25$ eV ($n_0 = 1.8 \times 10^{15}$ cm^{-3}): moving E_F 0.25 eV closer to E_c raises n_0 by about 1.6×10^4 times.
- **Repeated question:** the same question is 2(d)(ii) of the 2020–21 paper (extra reading: ../2020-21/2020-21-term-final.md).

7.(f) Silicon atoms, at a concentration of 7 × 10^{15} cm^{-3}, are added to gallium arsenide. Assume that the silicon atoms act as fully ionized dopant atoms and that 5 percent of the concentration added replaces gallium atoms and 95 percent replaces arsenic atoms. Let T = 300 K and GaAs has an intrinsic concentration of 1.8x10^6 cm $^{-3}$. i) Determine the donor and acceptor concentration. ii) Is the material n-type or p-type? iii) Calculate the electron and hole concentration. iv) Determine the position of the Fermi level with respect to E_{Fi}. Marks: 7

1. Theory

1. Gallium arsenide and its dopants (Core Idea)

GaAs is a **III-V compound**: every Ga atom (group III, three valence electrons) is bonded to four As atoms (group V, five valence electrons) and every As to four Ga. On average each atom brings four electrons to its bonds, as in Si. For GaAs at 300 K (Neamen Tables 4.1, 4.2): $E_g = 1.42$ eV, $N_c = 4.7 \times 10^{17}$ cm^{-3}, $N_v = 7.0 \times 10^{18}$ cm^{-3}, $n_i = 1.8 \times 10^6$ cm^{-3}.

An impurity is a donor or an acceptor depending on **which atom it replaces** (Neamen, Section 4.2.3):

- an atom with *one more* valence electron than the atom it replaces has a spare electron: a **donor**;
- an atom with **one fewer** has a missing bond electron: an **acceptor**.

2. Amphoteric impurities

Si is in group IV, with four valence electrons:

- on a **Ga site** (Ga has 3), Si has one electron too many: it acts as a **donor** and becomes Si$^+$;
- on an **As site** (As has 5), Si has one electron too few: it acts as an **acceptor** and becomes Si$^-$.

An impurity that can be either is called **amphoteric**. The same Si atoms can therefore produce donors and acceptors at the same time.

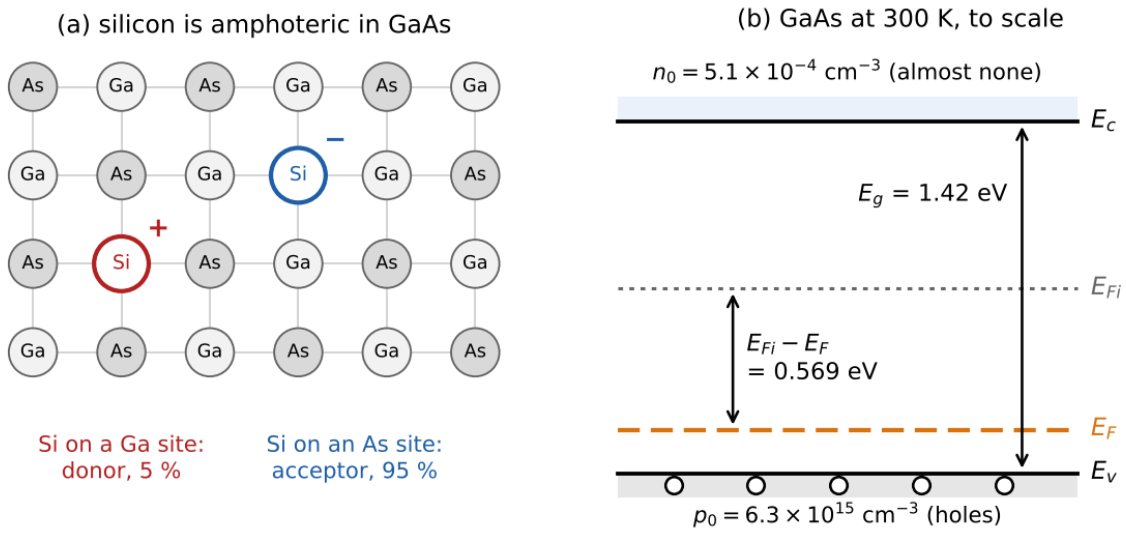


Figure 6: Silicon on a Ga site and on an As site in GaAs, and the band diagram of the resulting p-type GaAs

Reading the figure. Panel (a): a 2-D sketch of the GaAs lattice (light Ga, dark As). The Si with a red ring sits on a Ga site and is a donor (+); the Si with a blue ring sits on an As site and is an acceptor (−). Panel (b), to scale: the dashed orange E_F lies 0.569 eV below the dotted grey E_{Fi}, close to E_v; holes (open circles) are plentiful and electrons almost absent.

3. Compensation

When both donors and acceptors are present, the semiconductor is **compensated**. The electrons from the donors first fall into the acceptor states (lower energy), so they cancel each other; only the **difference** is left to make free carriers. If $N_a > N_d$ the material is p-type with $p_0 \approx N_a - N_d$; if $N_d > N_a$ it is n-type with $n_0 \approx N_d - N_a$.

4. Charge neutrality and the hole concentration (Neamen, Section 4.5)

A uniform crystal in equilibrium has no net charge: positive charges (holes, ionized donors) equal negative charges (electrons, ionized acceptors):

$$n_0 + N_a^- = p_0 + N_d^+ \qquad \text{(Neamen 4.56)}$$

With complete ionization ($N_a^- = N_a$, $N_d^+ = N_d$) and $n_0 = n_i^2/p_0$ (7(c), Step 11):

$$\frac{n_i^2}{p_0} + N_a = p_0 + N_d$$

Multiply by p_0 and collect terms:

$$p_0^2 - (N_a - N_d)\, p_0 - n_i^2 = 0$$

The quadratic formula, keeping the positive root (a concentration cannot be negative), gives (Neamen Eq. 4.62)

$$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$$

When $N_a - N_d \gg n_i$, the square root is $(N_a - N_d)/2$ and $p_0 \approx N_a - N_d$.

5. The Fermi level

From 7(c), Step 10, $p_0 = n_i \exp[(E_{Fi} - E_F)/kT]$. Taking the logarithm (Neamen Eq. 4.68):

$$E_{Fi} - E_F = kT \ln\left(\frac{p_0}{n_i}\right)$$

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In p-type material $p_0 > n_i$, so E_F is below E_{Fi}.

2. Answer

Given:

- Si added to GaAs: $N_{Si} = 7 \times 10^{15}$ cm^{-3}, fully ionized
- 5 % replace Ga atoms; 95 % replace As atoms
- $T = 300$ K, $kT = 0.0259$ eV; $n_i = 1.8 \times 10^6$ cm^{-3} (GaAs)

(i) Donor and acceptor concentrations

Si on a Ga site is a donor, Si on an As site is an acceptor (Theory 2):

$$N_d = 0.05 \times (7 \times 10^{15}) = 3.5 \times 10^{14} \text{ cm}^{-3}$$

$$N_a = 0.95 \times (7 \times 10^{15}) = 6.65 \times 10^{15} \text{ cm}^{-3}$$

$$\boxed{N_d = 3.5 \times 10^{14} \text{ cm}^{-3}, \qquad N_a = 6.65 \times 10^{15} \text{ cm}^{-3}}$$

(ii) n-type or p-type

$N_a > N_d$: after the donor electrons fill some of the acceptors, $N_a - N_d = 6.3 \times 10^{15}$ cm^{-3} acceptors are left to take electrons from the valence band.

$$\boxed{\text{The material is p-type (compensated)}}$$

(iii) Electron and hole concentrations

Holes (Theory 4, Neamen 4.62):

$$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2} = \frac{6.3 \times 10^{15}}{2} + \sqrt{(3.15 \times 10^{15})^2 + (1.8 \times 10^6)^2}$$

Since $n_i = 1.8 \times 10^6$ is 3×10^{-10} times $N_a - N_d$, the n_i^2 term changes nothing:

$$\boxed{p_0 = 6.3 \times 10^{15} \text{ cm}^{-3}}$$

Electrons (mass-action law):

$$n_0 = \frac{n_i^2}{p_0} = \frac{(1.8 \times 10^6)^2}{6.3 \times 10^{15}} = \frac{3.24 \times 10^{12}}{6.3 \times 10^{15}}$$

$$\boxed{n_0 = 5.14 \times 10^{-4} \text{ cm}^{-3}}$$

That is, about one electron in 2000 cm^3: practically no free electrons, because n_i of GaAs is so small.

(iv) Fermi level with respect to E_{Fi}

$$E_{Fi} - E_F = kT \ln\left(\frac{p_0}{n_i}\right) = 0.0259 \ln\left(\frac{6.3 \times 10^{15}}{1.8 \times 10^6}\right) = 0.0259 \ln\left(3.5 \times 10^9\right) = 0.0259 \times 21.976$$

$$\boxed{E_{Fi} - E_F = 0.569 \text{ eV}}$$

The Fermi level is 0.569 eV **below** the intrinsic Fermi level, as expected for p-type material. (Check: $n_0 = n_i \exp(-0.5692/0.0259) = 5.14 \times 10^{-4}$ cm^{-3}, as in (iii).)

Final Answer:

- (i) $N_d = 3.5 \times 10^{14}$ cm^{-3} (Si on Ga sites), $N_a = 6.65 \times 10^{15}$ cm^{-3} (Si on As sites)
- (ii) p-type, since $N_a > N_d$
- (iii) $p_0 = 6.3 \times 10^{15}$ cm^{-3}, $n_0 = 5.14 \times 10^{-4}$ cm^{-3}

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- (iv) E_F is 0.569 eV below E_{Fi}

3. Keyword Explanation

1. Amphoteric impurity

↩ back

An impurity that can act as a donor or an acceptor depending on the lattice site it takes (Si and Ge in GaAs).

2. Compensated semiconductor

↩ back

A semiconductor that has both donors and acceptors in the same region; the net doping $|N_d - N_a|$ sets the majority carrier density.

3. Charge neutrality

↩ back

$n_0 + N_a^- = p_0 + N_d^+$: a uniform sample in equilibrium carries no net charge.

4. Intrinsic Fermi level

↩ back

E_{Fi}, the Fermi level of undoped GaAs; it lies $\frac{1}{2}kT\ln(N_v/N_c) = 0.035$ eV above midgap (0.745 eV above E_v).

4. Notes and Remarks

- **Book source:** this is Neamen Problem 4.62 (Chapter 4 problems, p. 154), word for word, with the paper adding $n_i = 1.8 \times 10^6$ cm^{-3} (Neamen Table 4.2). The book prints no answer for it.
- **Position from the band edge:** $E_F - E_v = kT\ln(N_v/p_0) = 0.0259\ln(7.0 \times 10^{18}/6.3 \times 10^{15}) = 0.182$ eV. Through E_{Fi} (0.745 eV above E_v) one gets $0.745 - 0.569 = 0.176$ eV; the 6 meV difference again comes from N_c, N_v, E_g and n_i not being exactly consistent.
- **In practice:** Neamen notes that Si in GaAs is mostly a donor (it prefers the Ga site), so real Si-doped GaAs is n-type. The question assumes the opposite split to test the idea; with 95 % on Ga sites the material would be n-type with $n_0 = 6.3 \times 10^{15}$ cm^{-3} and E_F 0.569 eV above E_{Fi}.
- **Freeze-out is not a concern:** donor and acceptor ionization energies in GaAs are only about 0.006 eV (donors, Si among them) to 0.04 eV (acceptors; Si on an As site 0.0345 eV) (Neamen Table 4.4), so "fully ionized" is a safe assumption at 300 K.

5. Final Concise Answer

(a) The Fermi-Dirac distribution $f_F(E) = 1/[1 + \exp((E - E_F)/kT)]$ is the probability that an allowed quantum state at energy E is occupied by an electron in thermal equilibrium (particles that cannot be told apart, at most one per state). At 0 K it is a step (full below E_F, empty above); for $T > 0$ it is 1/2 at E_F and changes over a few kT; $1 - f_F$ is the probability of a hole, and for $E - E_F \gg kT$, $f_F \approx e^{-(E-E_F)/kT}$.

(b) Ionization energy is the energy needed to free the extra electron of a donor ($E_c - E_d$) or to put a valence electron into an acceptor ($E_a - E_v$), leaving a fixed ion and a free carrier. The hydrogen model gives $13.6\,(m^*/m_0)/\epsilon_r^2 = 25.8$ meV for Si; measured values are 0.045 eV (P, B) and 0.05 eV (As), much smaller than E_g and close to kT, so dopants are fully ionized at room temperature.

(c) $n(E) = g_c f_F$ integrated over the conduction band with the Boltzmann approximation and $\int_0^\infty \eta^{1/2} e^{-\eta} d\eta = \sqrt{\pi}/2$ gives $n_0 = N_c \exp[-(E_c - E_F)/kT]$, $N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$; likewise $p_0 = N_v \exp[-(E_F - E_v)/kT]$. With E_{Fi} as reference, $n_0 = n_i \exp[(E_F - E_{Fi})/kT]$ and $p_0 = n_i \exp[-(E_F - E_{Fi})/kT]$, and $n_0 p_0 = n_i^2$. n-type: $E_F > E_{Fi}$, $E_F - E_{Fi} = kT\ln(N_d/n_i)$; p-type: $E_{Fi} - E_F = kT\ln(N_a/n_i)$.

(d) $f_F = 0.10$ gives $e^{(E-E_F)/kT} = 9$, so $E - E_F = kT\ln 9 = 2.197\,kT \approx 2.2\,kT$ (0.0569 eV at 300 K).

(e) $n_0 = 2.8 \times 10^{19} \exp(-0.5/0.0259) = 2.8 \times 10^{19} \times 4.13 \times 10^{-9} = 1.16 \times 10^{11}$ cm^{-3}.

(f) (i) $N_d = 3.5 \times 10^{14}$ cm^{-3}, $N_a = 6.65 \times 10^{15}$ cm^{-3}. (ii) p-type. (iii) $p_0 = 6.3 \times 10^{15}$ cm^{-3}, $n_0 = n_i^2/p_0 = 5.14 \times 10^{-4}$ cm^{-3}. (iv) $E_{Fi} - E_F = 0.0259\ln(3.5 \times 10^9) = 0.569$ eV (below E_{Fi}).

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FORMULA SHEET

EEE421

Solid State Devices

Formula Sheet

Session 2021–22 term test *Every formula behind Tutorial 01 (Q7); the final examination is not yet held*

TOPICS

1 The Fermi-Dirac function — 5 formulas

2 Dopants and ionization energy — 4 formulas

3 Carriers in equilibrium — 9 formulas

4 Compensated material — 4 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

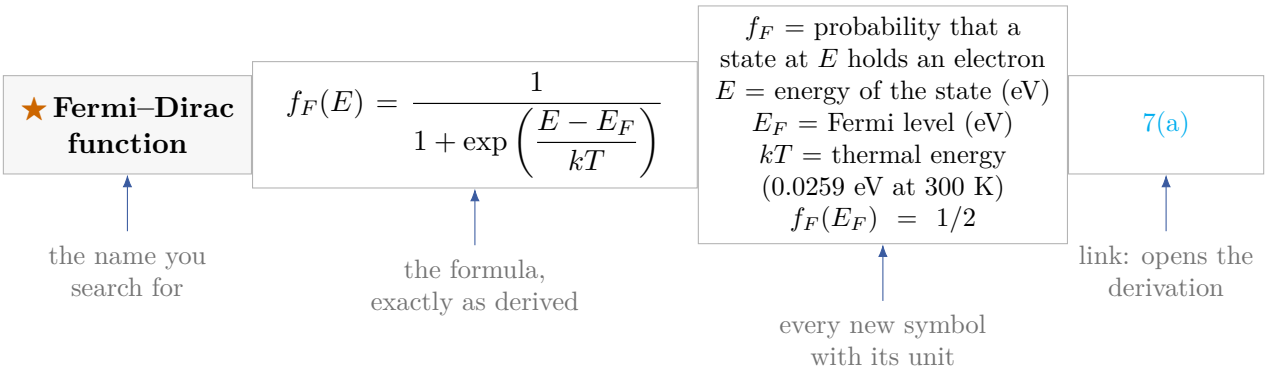
Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 The Fermi-Dirac function **4**

2 Dopants and ionization energy **5**

3 Carriers in equilibrium **5**

4 Compensated material **7**

5 Key formulas **8**

6 Symbol index **9**

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Conventions (Neamen, 4th ed.): e is the magnitude of the electron charge, $kT = 0.0259$ eV at 300 K, $k = 8.62\times10^{-5}$ eV/K. Si at 300 K: $N_c = 2.8 \times 10^{19}$ cm^{-3}, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $E_g = 1.12$ eV, $\epsilon_r = 11.7$. GaAs at 300 K: $N_c = 4.7 \times 10^{17}$ cm^{-3}, $N_v = 7.0 \times 10^{18}$ cm^{-3}, $n_i = 1.8 \times 10^{6}$ cm^{-3}, $E_g = 1.42$ eV. References point to the solutions of the 2021–22 term test.

Columns: Title · Equation · Notation · Ref. (reference).

1. The Fermi-Dirac function:

Title	Equation	Notation	Ref.
★ **Fermi–Dirac function**	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$	f_F = probability that a state at E holds an electron E = energy of the state (eV) E_F = Fermi level (eV) kT = thermal energy (0.0259 eV at 300 K) $f_F(E_F) = 1/2$	7(a)
Ways to place electrons	$W_i = \dfrac{g_i!}{N_i!\,(g_i - N_i)!}$	W_i = number of ways to place the electrons in level i g_i = number of states in level i N_i = number of electrons in level i	7(a)
Empty-state probability	$1 - f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E_F-E}{kT}\right)}$	symmetric with f_F about E_F	7(a)
★ **Boltzmann approximation**	$f_F(E) \approx \exp\left[-\dfrac{E-E_F}{kT}\right]$	valid for $E - E_F \gtrsim 3kT$ (5 % error at $3kT$)	7(a)
★ **Energy for a given probability**	$E - E_F = kT \ln\left(\dfrac{1}{f_F} - 1\right)$	$f_F = 0.1$: $kT \ln 9 = 2.197\,kT$ (0.0569 eV)	7(d)

Table 1: The Fermi-Dirac function

2. Dopants and ionization energy:

Title	Equation	Notation	Ref.
Bohr orbit radius	$\frac{r_n}{a_0} = n^2 \epsilon_r \frac{m_0}{m^*}$	r_n = radius of the n-th orbit (Å) a_0 = Bohr radius (0.53 Å) n = orbit number (1 for the ground state) ϵ_r = relative permittivity (11.7 for Si) m_0 = free-electron mass (kg) m^* = conductivity effective mass (kg) Si: $r_1 = 45a_0 = 23.8$ Å	7(b)
★ **Donor energy (hydrogen model)**	$E = -\frac{m^* e^4}{2(n\hbar)^2(4\pi\epsilon)^2}$	e = magnitude of the electron charge (1.6×10^{-19} C) $\hbar$ = reduced Planck constant, $h/2\pi$ (J s) ϵ = permittivity of the semiconductor, $\epsilon_r\epsilon_0$ (F/m) E = energy of the bound electron in orbit n (J)	7(b)
★ **Ionization energy**	$E_{ion} = 13.6 \text{ eV} \times \frac{m^*/m_0}{\epsilon_r^2}$	E_{ion} = donor ionization energy (eV) Si ($m^* = 0.26m_0$): 25.8 meV; measured P 0.045, As 0.05 eV	7(b)
Donor and acceptor	donor: $E_c - E_d$; acceptor: $E_a - E_v$	E_c = conduction-band edge (eV) E_d = donor energy level (eV) E_a = acceptor energy level (eV) E_v = valence-band edge (eV) Si in GaAs: donor on a Ga site, acceptor on an As site	7(b), 7(f)

Table 2: Dopants and ionization energy

3. Carriers in equilibrium:

Title	Equation	Notation	Ref.
Carriers per energy	$n(E) = g_c(E) f_F(E)$, $p(E) = g_v(E)[1 - f_F(E)]$	g_c, g_v = densities of states in the conduction and valence bands ($\text{cm}^{-3}\text{eV}^{-1}$) $n(E)$ = electrons per unit energy ($\text{cm}^{-3}\text{eV}^{-1}$) $p(E)$ = holes per unit energy ($\text{cm}^{-3}\text{eV}^{-1}$) f_F = probability that a state at E holds an electron E = energy of the state (eV) Neamen 4.1, 4.2	7(c)
★ **Density of states**	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	m_n^* = electron effective mass (kg) h = Planck's constant (6.625×10^{-34} J s) $g_v(E)$: the same with m_p^* and $\sqrt{E_v - E}$ m_p^* = hole effective mass (kg) E_c = conduction-band edge (eV)	7(c)
Integral used	$\int_0^\infty \eta^{1/2} e^{-\eta}\, d\eta = \frac{\sqrt{\pi}}{2}$	η = energy above the band edge in units of kT, $(E - E_c)/kT$	7(c)

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Title	Equation	Notation	Ref.
★ **Electron concentration**	$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	n_0 = thermal-equilibrium electron concentration (cm^{-3}) N_c = effective density of states in the conduction band, $2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$ (cm^{-3}) E_F = Fermi level (eV) kT = thermal energy (0.0259 eV at 300 K) $E_c - E_F = 0.5$ eV in Si: $n_0 = 1.16 \times 10^{11}$ cm^{-3}	7(c), 7(e)
★ **Hole concentration**	$p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	p_0 = thermal-equilibrium hole concentration (cm^{-3}) N_v = effective density of states in the valence band, $2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$ (cm^{-3}) E_v = valence-band edge (eV)	7(c)
★ **From the intrinsic level**	$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$, $p_0 = n_i \exp\left[-\frac{E_F - E_{Fi}}{kT}\right]$	n_i = intrinsic carrier concentration (cm^{-3}) E_{Fi} = intrinsic Fermi level (eV)	7(c)
★ **Mass-action law**	$n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$	E_g = band gap energy (eV) non-degenerate, equilibrium	7(c)
Fermi level, n-type	$E_F - E_{Fi} = kT \ln\frac{n_0}{n_i}$, $E_c - E_F = kT \ln\frac{N_c}{n_0}$	N_d = donor concentration (cm^{-3}) $n_0 \approx N_d$ (complete ionization)	7(c)
★ **Fermi level, p-type**	$E_{Fi} - E_F = kT \ln\frac{p_0}{n_i}$, $E_F - E_v = kT \ln\frac{N_v}{p_0}$	7(f): 0.569 eV below E_{Fi}	7(c), 7(f)

Table 3: Carriers in equilibrium

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4. Compensated material:

Title	Equation	Notation	Ref.
★ **Charge neutrality**	$n_0 + N_a^- = p_0 + N_d^+$	N_a = acceptor concentration (cm^{-3}) N_a^- = ionized acceptor concentration (cm^{-3}) N_d^+ = ionized donor concentration (cm^{-3}) n_0 = thermal-equilibrium electron concentration (cm^{-3}) p_0 = thermal-equilibrium hole concentration (cm^{-3}) complete ionization: $N_a^- = N_a$, $N_d^+ = N_d$	7(f)
★ **Holes, compensated**	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$	n_i = intrinsic carrier concentration (cm^{-3}) N_d = donor concentration (cm^{-3}) $p_0 \approx N_a - N_d$ when $N_a - N_d \gg n_i$	7(f)
Amphoteric split	$N_d = x_{Ga} N_{Si}$, $N_a = x_{As} N_{Si}$	N_{Si} = silicon atoms added to the GaAs (cm^{-3}) x_{Ga} = fraction of Si on Ga sites (donors) x_{As} = fraction of Si on As sites (acceptors) 7(f): $N_d = 3.5 \times 10^{14}$, $N_a = 6.65 \times 10^{15}$ cm^{-3}	7(f)
Minority carriers	$n_0 = \frac{n_i^2}{p_0}$	7(f): 5.14×10^{-4} cm^{-3}	7(f)

Table 4: Compensated material

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5. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Fermi–Dirac function	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$
Boltzmann approximation	$f_F(E) \approx \exp\left[-\dfrac{E-E_F}{kT}\right]$
Energy for a given probability	$E - E_F = kT \ln\left(\dfrac{1}{f_F} - 1\right)$
Donor energy (hydrogen model)	$E = -\dfrac{m^* e^4}{2(n\hbar)^2(4\pi\epsilon)^2}$
Ionization energy	$E_{ion} = 13.6 \text{ eV} \times \dfrac{m^*/m_0}{\epsilon_r^2}$
Density of states	$g_c(E) = \dfrac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E-E_c}$
Electron concentration	$n_0 = N_c \exp\left[-\dfrac{E_c - E_F}{kT}\right]$
Hole concentration	$p_0 = N_v \exp\left[-\dfrac{E_F - E_v}{kT}\right]$
From the intrinsic level	$n_0 = n_i \exp\left[\dfrac{E_F - E_{Fi}}{kT}\right],\ p_0 = n_i \exp\left[-\dfrac{E_F - E_{Fi}}{kT}\right]$
Mass-action law	$n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\dfrac{E_g}{kT}\right)$
Fermi level, p-type	$E_{Fi} - E_F = kT \ln\dfrac{p_0}{n_i},\ E_F - E_v = kT \ln\dfrac{N_v}{p_0}$
Charge neutrality	$n_0 + N_a^- = p_0 + N_d^+$
Holes, compensated	$p_0 = \dfrac{N_a - N_d}{2} + \sqrt{\left(\dfrac{N_a - N_d}{2}\right)^2 + n_i^2}$

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6. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
f_F	probability that a state at E holds an electron	Fermi–Dirac function
E	energy of the state (eV)	Fermi–Dirac function
E_F	Fermi level (eV)	Fermi–Dirac function
kT	thermal energy (0.0259 eV at 300 K)	Fermi–Dirac function
W_i	number of ways to place the electrons in level i	Ways to place electrons
g_i	number of states in level i	Ways to place electrons
N_i	number of electrons in level i	Ways to place electrons
r_n	radius of the n-th orbit (Å)	Bohr orbit radius
a_0	Bohr radius (0.53 Å)	Bohr orbit radius
n	orbit number (1 for the ground state)	Bohr orbit radius
ϵ_r	relative permittivity (11.7 for Si)	Bohr orbit radius
m_0	free-electron mass (kg)	Bohr orbit radius
m^*	conductivity effective mass (kg)	Bohr orbit radius
e	magnitude of the electron charge (1.6×10^{-19} C)	Donor energy (hydrogen model)
$\hbar$	reduced Planck constant, $h/2\pi$ (J s)	Donor energy (hydrogen model)
ϵ	permittivity of the semiconductor, $\epsilon_r\epsilon_0$ (F/m)	Donor energy (hydrogen model)
E	energy of the bound electron in orbit n (J)	Donor energy (hydrogen model)
E_{ion}	donor ionization energy (eV)	Ionization energy
E_c	conduction-band edge (eV)	Donor and acceptor
E_d	donor energy level (eV)	Donor and acceptor
E_a	acceptor energy level (eV)	Donor and acceptor
E_v	valence-band edge (eV)	Donor and acceptor
g_c, g_v	densities of states in the conduction and valence bands ($\mathrm{cm^{-3}eV^{-1}}$)	Carriers per energy
$n(E)$	electrons per unit energy ($\mathrm{cm^{-3}eV^{-1}}$)	Carriers per energy
$p(E)$	holes per unit energy ($\mathrm{cm^{-3}eV^{-1}}$)	Carriers per energy
m_n^*	electron effective mass (kg)	Density of states
h	Planck's constant (6.625×10^{-34} J s)	Density of states
m_p^*	hole effective mass (kg)	Density of states
η	energy above the band edge in units of kT, $(E - E_c)/kT$	Integral used
n_0	thermal-equilibrium electron concentration ($\mathrm{cm^{-3}}$)	Electron concentration
N_c	effective density of states in the conduction band, $2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$ ($\mathrm{cm^{-3}}$)	Electron concentration

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Symbol	Meaning	First used in
p_0	thermal-equilibrium hole concentration (cm^{-3})	Hole concentration
N_v	effective density of states in the valence band, $2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$ (cm^{-3})	Hole concentration
n_i	intrinsic carrier concentration (cm^{-3})	From the intrinsic level
E_{Fi}	intrinsic Fermi level (eV)	From the intrinsic level
E_g	band gap energy (eV)	Mass-action law
N_d	donor concentration (cm^{-3})	Fermi level, n-type
N_a	acceptor concentration (cm^{-3})	Charge neutrality
N_a^-	ionized acceptor concentration (cm^{-3})	Charge neutrality
N_d^+	ionized donor concentration (cm^{-3})	Charge neutrality
N_{Si}	silicon atoms added to the GaAs (cm^{-3})	Amphoteric split
x_{Ga}	fraction of Si on Ga sites (donors)	Amphoteric split
x_{As}	fraction of Si on As sites (acceptors)	Amphoteric split

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